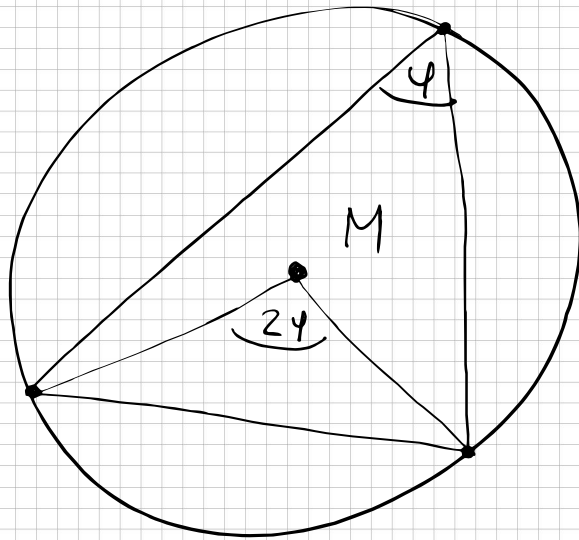
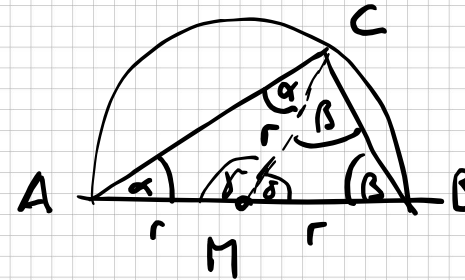


Peripheriewinkelsatz



Fall 1:



$$2\alpha + \gamma = 180^\circ$$

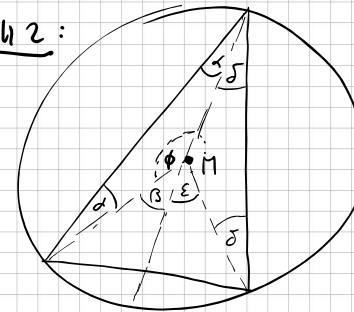
$$2\beta + \gamma = 180^\circ$$

$$\gamma + \gamma = 180^\circ$$

$$\hookrightarrow 2\alpha + 2\beta + \overset{180^\circ}{\gamma} = 360^\circ$$

$$\Leftrightarrow \alpha + \beta = 90^\circ$$

Fall 2:



$$2\alpha + \phi = 180^\circ$$

$$2\delta + \psi = 180^\circ$$

$$\phi + \beta = 180^\circ$$

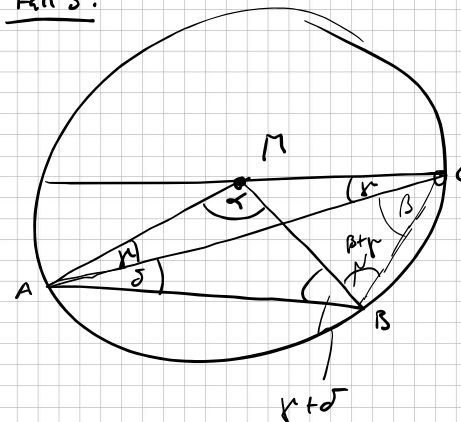
$$\psi + \epsilon = 180^\circ$$

$$\Rightarrow \beta = 2\alpha$$

$$\Rightarrow 2\delta = \epsilon$$

$$\hookrightarrow \beta + \epsilon = 2\alpha + 2\delta = 2(\alpha + \delta)$$

Fall 3:



$$ABM: \alpha + 2\gamma + 2\delta = 180^\circ$$

$$ABC: 2\delta + 2\gamma + 2\beta = 180^\circ$$

$$\Leftrightarrow 2\delta + 2\gamma = 180^\circ - 2\beta$$

$$\hookrightarrow \alpha = 2\beta$$

□